



Spring 2026 MEMS 3420 Heat Transfer

Heat Transfer COMSOL Final Project

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Introduction

During winter, when temperatures range from -5°C to 10°C and conditions can be unpredictable, heated blankets provide a convenient and reliable source of immediate thermal comfort without relying solely on body heat. Despite being a common household item, heated blankets rely on key heat transfer mechanisms that are often overlooked. Investigating how effectively these systems transfer heat, and how long it takes to heat, through conduction and convection offers valuable insight into their performance and highlights opportunities for improving efficiency. The goal for this analysis is to determine the perfect time to start heating our blanket before we are ready to go to bed. Our goal temperature is 40°C and the blanket is starting at an ambient temperature of 22°C . Heat generation is applied to a third of the blanket separated off into 8 different sections.

Pre-analysis

Below lists general equations, parameters, and boundary conditions for this problem.

0.1 Parameter Values.

$$L = 0.02 \text{ m}$$

$$\rho = 100 \text{ kg/m}^3 \text{ [1]}$$

$$c_p = 950 \text{ J/(kg} \cdot \text{K)} \text{ [2]}$$

$$h_{\text{fleece}} = 0.5 \text{ W/(m}^2 \cdot \text{K)} \text{ [3]}$$

$$k_{\text{fleece}} = 0.004 \text{ W/(m} \cdot \text{K)}$$

$$T_{\infty} = 22^{\circ}\text{C} \text{ [4]}$$

$$T = 40^{\circ}\text{C}$$

$$h_{\text{air}} = 0.5 \text{ W/(m}^2 \cdot \text{K)}$$

0.2 General Equations.

$$\rho c_p \frac{\partial T}{\partial t} = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + \dot{q}, \quad \text{with a range of } 0 \leq x \leq 0.02 \text{ m, } 0 \leq y \leq 2 \text{ m.} \quad (1)$$

0.3 Boundary Conditions.

Bottom surface (insulated, $x=0$)

$$\left. \frac{\partial T}{\partial x} \right|_{x=0} = 0 \quad (2)$$

Top surface (convection, x=0.02)

$$-k \left. \frac{\partial T}{\partial x} \right|_{x=0.02} = h (T - T_{\infty}) \quad (3)$$

Side boundaries (adiabatic, y=0,2)

$$\left. \frac{\partial T}{\partial y} \right|_{y=0,2} = 0 \quad (4)$$

Target Condition

$$T(x, y, t) = 40^{\circ}\text{C} \quad (5)$$

0.4 Analytical Solution.

To attempt to analytically solve, a full 2D analytical solution with generation and mixed boundary conditions isn't something to realistically write out, as it becomes a messy infinite series.

However, the system at hand is strongly dominated in the width direction (2 cm vs 2 m), so a 1D transient analytical solution is a justified approach. So equation 6, becomes

$$\rho c_p \frac{\partial T}{\partial t} = k \left(\frac{\partial^2 T}{\partial x^2} \right) + \dot{q} \quad (6)$$

To estimate the time required for the heated blanket to reach 40°C, a lumped capacitance approach is used. This is justified due to the small thickness of the blanket relative to its width.

Using equation 3, and applying an energy balance with internal heat generation:

$$\rho c_p \frac{dT}{dt} = \dot{q} - \frac{hA}{V} (T - T_{\infty}) \quad (7)$$

where the height and width terms cancel between the area and volume expressions, reducing equation 7 to equation 8.

$$\rho c_p \frac{dT}{dt} = \dot{q} - \frac{h}{L} (T - T_{\infty}) \quad (8)$$

The solution to this first-order differential equation is:

$$T(t) = T_{\infty} + \frac{\dot{q}L}{h} \left(1 - e^{-\frac{h}{\rho c_p L} t} \right) \quad (9)$$

Rearranging equation 9 to solve for time, we get equation 10

$$t = -\tau \ln \left(1 - \frac{(T - T_{\infty})h}{\dot{q}L} \right) \quad (10)$$

where the time constant is:

$$\tau = \frac{\rho c_p L}{h} \quad (11)$$

Solving for t using equation 10,

$$t \approx 847 \text{ s} \approx 8.5 \times 10^2 \text{ s} \approx 14 \text{ minutes} \quad (12)$$

0.5 Numerical Result Comparison.

The wattage of a heated blanket greatly varies the time it takes to heat a blanket. Despite this, heated blankets, in general, typically take 10-30 minutes to heat up [5]. Because heated blankets typically have 50-150 W [6], and our blanket has 100 W, we would expect a heating time of around 20 minutes. Although the analytical solution predicts a slightly shorter time, it remains within the expected range for reaching a comfortable temperature of 40°C.

1 Geometry, Mesh, and Setup in COMSOL

1.6 Geometry. In order to determine the heat transfer characteristics, the 2D heated blanket system was modeled as a 2D rectangular domain in COMSOL Multiphysics. The dimension chosen for the rectangle was a width of 2.0 m and a thickness of 0.02 m.

In order to evaluate the thermal gradients of the heated blanket, the 2.0 m width of the model was sectioned into 24 domains. From the total 24 domains, 8 domains were used as heating domains and were evenly spaced across the model. The other 16 regions represent the unheated fleece that the temperature is being distributed. The 300 W of total power was distributed evenly across the heat generator domains resulting in 37.5 W for each wire heating region. By making the separate segments

of the rectangular domain, the heat conducts outward from each wire region, which spreading laterally and vertically throughout the unheated fleece segments. This allows for the simulation to be a realistic 2D thermal distribution instead of a 1D heat flow system. In this modeling, no lines of symmetry were used in order to visual the whole cross-section of the model.

Figure 1 Depicts the Geometry for a Heated Blanket.

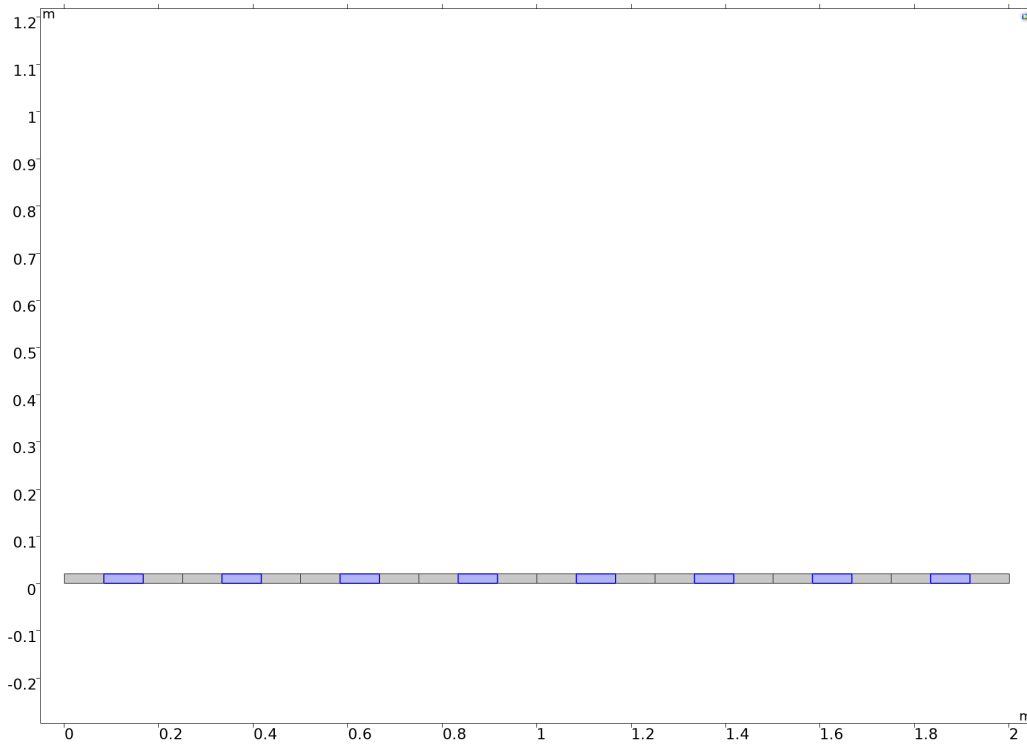


Figure 1 Heated Blanked COMSOL Geometry

1.7 Mesh. When creating a mesh for the rectangular domain, a physics-controlled mesh was applied. The blanket has a high aspect ratio since with has a 2.0 m width and a 0.02 m thickness. Free triangular elements were used for the mesh to determine the thermal gradients near the boundaries of the inner region in order to accurately map the mesh.

In order to determine that the results of the heated blanked is independent of the mesh size, a mesh refinement study was run. The simulation was first run using a "Normal" element size. After, the element size was change to "Fine" and "Coarse" mesh resolutions. Each time these mesh resolutions were run, the temperature contours reached identical patterns. Additionally, the heat diffuses outward from the wires in the same pattern across the meshes. The unheated regions encountered a surface temperature of around 40 ° Celsius. These results provide the data to show that the numerical results

are independent of the mesh sizing and that the study has gained mesh independence. The "Normal" mesh resolutions was selected in order to depict the thermal boundary layers in an efficient manner.

Figure 2 Depicts the "Normal" Mesh for a Heated Blanket.



Figure 2 Heated Blanked COMSOL "Normal" Mesh

1.8 COMSOL Setup. In order to understand how the heated blanked can provide a consistent source of immediate thermal comfort, a selective process was used to enact the environment in COMSOL.

We created a fleece material that had a thermal conductivity of $0.004 \frac{W}{m \cdot K}$. The density is $20 \frac{kg}{m^3}$. The heat capacity at a constant pressure is 950 (c_p). We used the built in property values from COMSOL for our air parameters. Transient state was used.

We simulated the blanket as a two-dimensional side-view model with a length of 2 m and a thickness of 20 mm. The rectangular domain was divided into 24 equal sections. To approximate the heating wires, heat sources were applied to 8 sections: sections 2, 5, 8, 11, 14, 17, 20, and 23. In other words, every third section contained a heat source. Each heat source generated 37.5 W, giving a total heating power of 300 W. The initial temperature of the whole blanket was set to the ambient

temperature of 22 °C.

The left, bottom, and right boundaries of the rectangle were assumed to be insulated. On the top boundary, a heat flux boundary condition was applied to represent external forced convection. The surrounding air temperature was set to 22 °C, and the air velocity was set to 0.25 m/s.

To compare the effect of mesh resolution, physics-controlled meshes with coarse, normal, and fine settings were tested separately. For each mesh setting, the temperature distribution and the time-dependent average temperature of the top surface were plotted.

The simulation time was defined as $\text{range}(0, 1, 1500)$, corresponding to a time step of 1 s and a total simulation time of 1500 s, or 25 min. The purpose of the simulation was to determine how long it would take for the average top surface temperature to reach 40 °C.

2 Numerical Results

2.1 Mesh Independence. In order to confirm the accuracy of the numerical solutions, a mesh refinement study was conducted. The mesh refinement study calculated the differences between a Coarse, Normal, and Fine mesh configurations. The surface temperatures were conducted above the central heating elements at $t = 1500$ s. The maximum surface temperature for the Normal mesh was found to be 40.09 °C, while the Fine mesh yielded the same temperature outcome. Since the surface temperatures were identical, the model solution for the heated blanket is mesh independent. This ensures computational accuracy.

2.2 Transient Temperature Distribution. The COMSOL modeling was primarily used to visualize the 2D thermal gradients found across the blanket from the heat generating regions. Figure 3 represents the temperature contour map of the blanket cross-section at $t = 1500$ s.

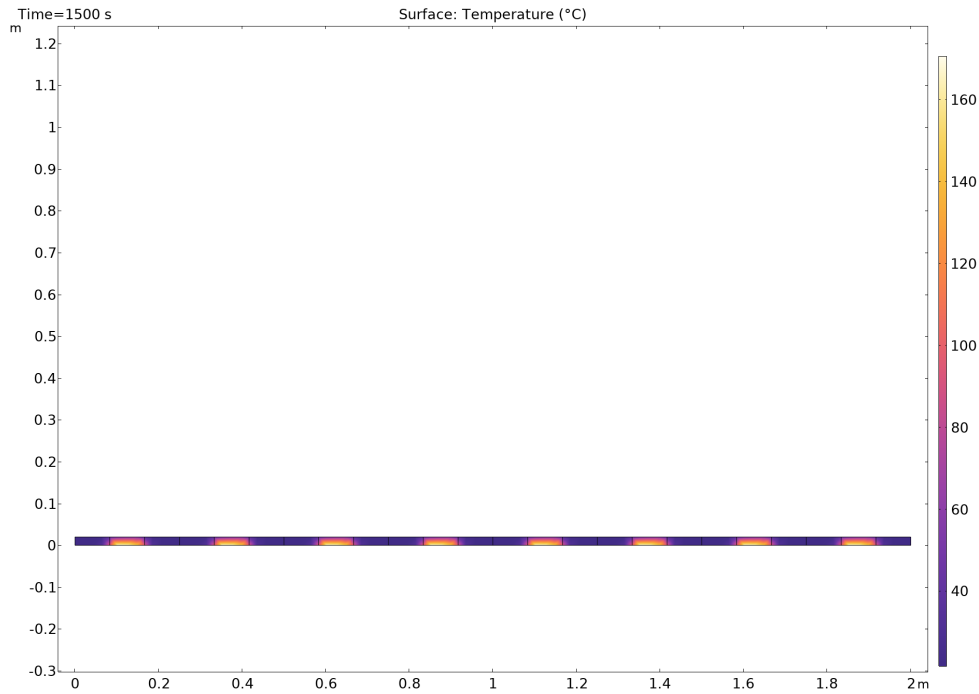


Figure 3 Temperature Distribution over Normal Mesh Modeling

The temperature distribution across the heated blanket is highly non uniform which is to be expected. The maximum temperatures are found within the heating elements and peak at around 160°C. From the heating sources, the thermal energy conducts outward through the fleece regions. The regions of fleece are located in between the wiring and each have much lower temperatures at the desired 40°C range. The varying patterns of heat distribution represent the necessity for a 2D model given the heated blanket scenario. A 1D model would be lacking in regards to depicting the high and low temperature zones as well as the lateral conduction resistance present from the fleece material regions.

2.3 Temperature Progression with Time. Figure 4 shows the transient temperature at the top of the blanket as it progresses with time. It can be seen to start at the ambient temperature of 22°C and gradually increase to the goal temperature of 40°C. The graph displays an exponential growth and this occurs because as time passes the convective heat loss from the surrounding becomes more effective causing a slower increase. The plot indicates that it took 1500 seconds (25 minutes) for the blanket to reach the goal temperature. This is within the expected range of 10-30 minutes however it is higher than the analytic results. This could be a result from the setup that we decided to use being

2D while the analytical results is calculated in 1D.

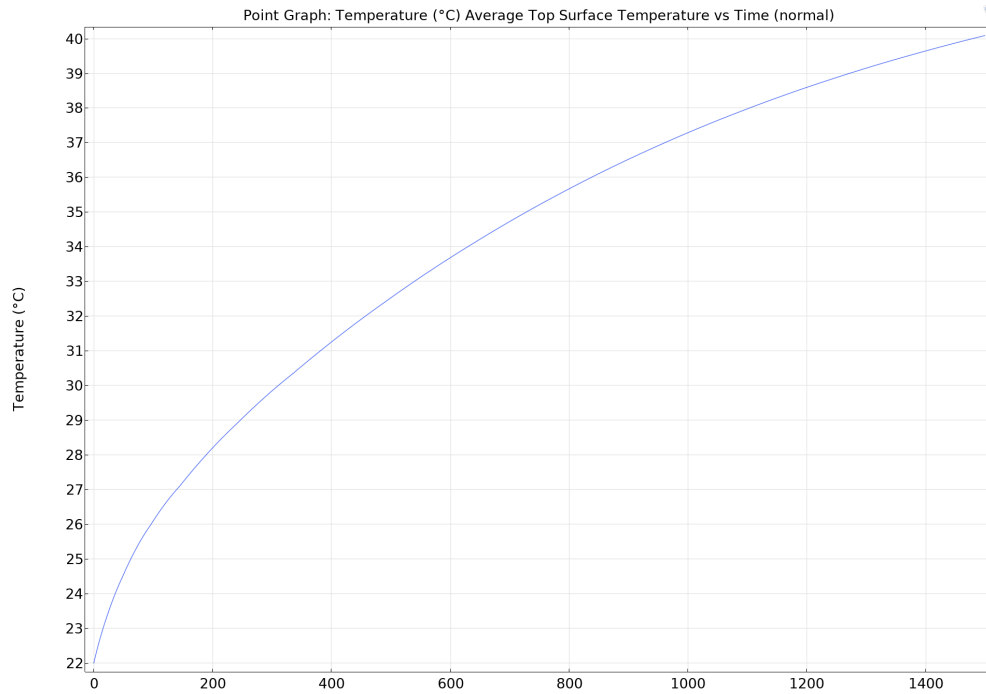


Figure 4 Temperature over the Top Surface with Time

2.4 Comparison to Analytical Theory. In order to check the computation model, the results were compared to the 1D lumped capacitance analytical solution. The analytical model was found by using a uniform internal heat generation and it predicted that the blanket would need 847 seconds to reach the targeted comfort temperature of 40°C.

When running the COMSOL simulation, the average surface temperature was evaluated across the top boundary layer of the heated blanket at $t = 847$ s. When running the simulation on COMSOL, the average surface temperature reached 31.16° C. The differences in temperatures from the analytical and computational approaches highlight the inaccuracies found using the lumped capacitance assumption for the heated blanket system. Since fleece is acting as an insulator, this means that the real world application will have an internal temperature gradient. From the computational study, the heat conducts away from the wires while the outer surface warms slowly to reach the desired temperature. The lower average surface temperature found using COMSOL helps depict why the computational model is a more realistic approach to finding the heat dynamics within the heated blanket system.

Conclusion

The study was able to successfully model the 2D heat transfer properties of an electrically heated blanket. By using the 2D setup with 24 regions, the model depicted the thermal gradient distributions across the blanket from resistive heating. The mesh refinement study showed how the numerical solutions of the study were independent of element sizing from the various meshes, which was used to verify the computational analysis.

One limitation that was found within our study was the model's assumption that the 300 W of total power would continuously input over the entire time of the study. The constant input resulted in the heating elements peaking at 160 ° Celsius. This peak temperature could be hazardous to the fleece material in a real world application. In reality, electrically heated blankets use thermostatic controllers that ensure the temperatures are distributed more evenly at safer temperatures over the time required to heat the blanket. It is important to note that the study conducted from our group represents the maximum output application instead of a regulated scenario seen in actual heated blankets.

Initially, the 1D pre-analysis used lumped capacitance to predict a heating time of 14 minutes. The 1D method assumes an infinite internal thermal conductivity. Since the fleece material acts as an insulator within the heated blanket, a thermal gradient will exist within the model which makes the 1D model less physically feasible. The 2D model was used to capture the thermal gradients at a more realistic approach.

References

- [1] n.a., 2025, “Fleece Weight,” tvfinc, accessed n.d., <https://www.tvfinc.com/article/types-of-fleece/#:~:text=Lightweight%20Fleece%20%E2%80%93%20100%20g/m2,is%20ideal%20for%20outer%20layers>.
- [2] Verma, P., 2025, “Specific Heat Capacity of Textile Fibers,” kamlatech, accessed 2022, <https://blog.kamlatech.in/specific-heat-capacity-of-textile-fibers/#:~:text=December%2028%2C%202022%20%C2%B7%202%20min,Why%20you%20ask?>
- [3] de Dear 1, R. J., May 1997, “Convective and radiative heat transfer coefficients for individual human body segments,” National Library of Medicine, accessed 2023, <https://pubmed.ncbi.nlm.nih.gov/9195861/>
- [4] n.a., 2025, “What Is the Normal Room Temperature?” AirZone, accessed n.d., https://www.airzonecontrol.com/na/en/blog/normal-room-temperature/#:~:text=At%20Home%20*%20For%20living%20areas%2C%20the,as%20it%20promotes%20better%20rest%20and%20relaxation.
- [5] de Dear 1, R. J., May 1997, “How Long Does an Electric Blanket Take to Heat Up: Quick Guide,” Puffy, accessed 2023, https://puffy.com/blogs/best-sleep/how-long-does-an-electric-blanket-take-to-heat-up?srltid=AfmBOopUOdrSwbP7nqGHNLU6kPfVAap1CbbXpA2kMmL7nU3DL0zxnJ_8&referrer=https%3A%2F%2Fwww.google.com%2F
- [6] n.a., 2025, “How Much Electricity Does a Heated Blanket Use? Complete Energy Usage Guide,” SolarTech, accessed 2025, <https://solartechonline.com/blog/how-much-electricity-heated-blanket-use/#:~:text=Twin%20size:%2060%2D100%20watts,King%20size:%20150%2D200%20watts>